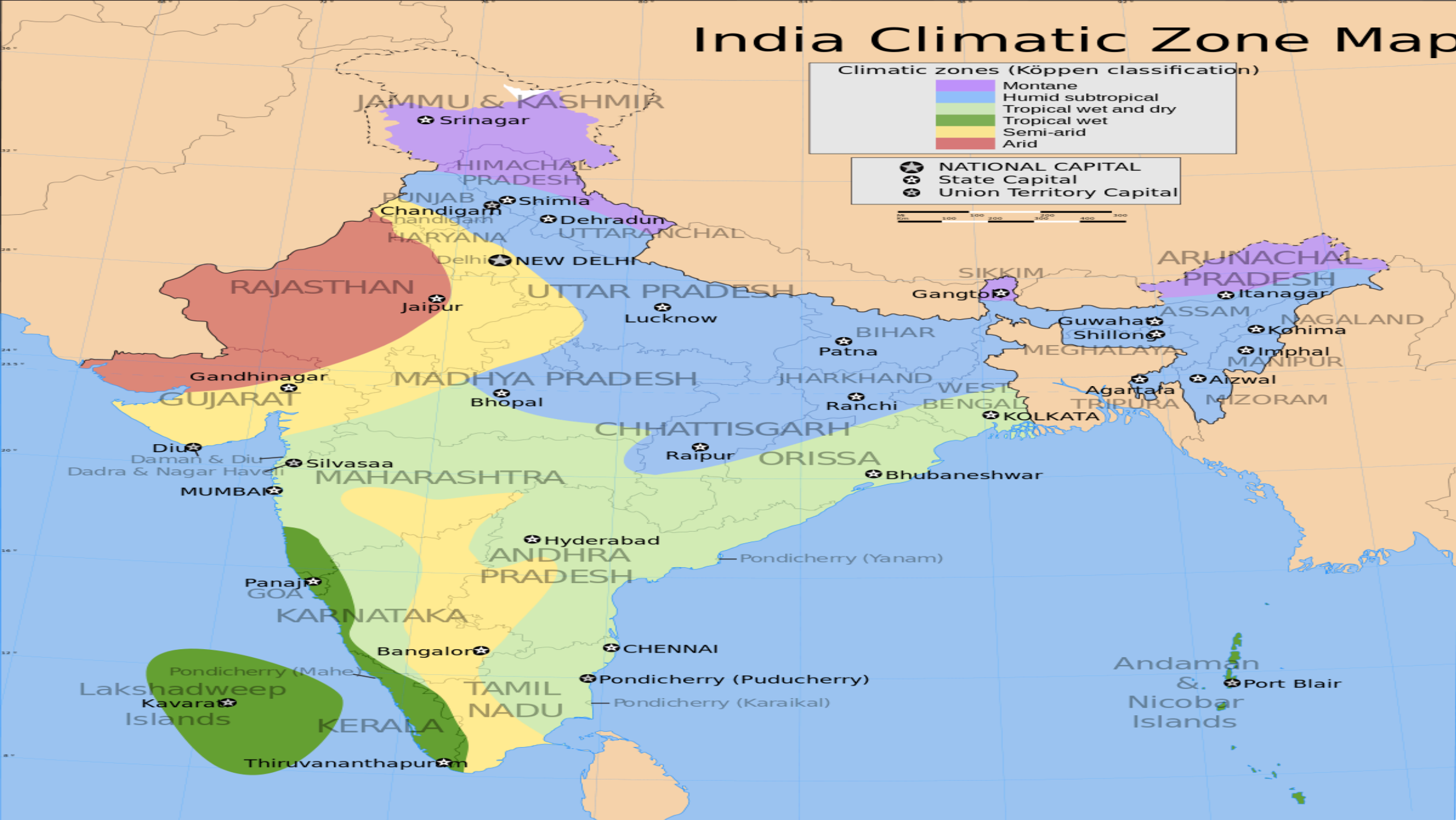
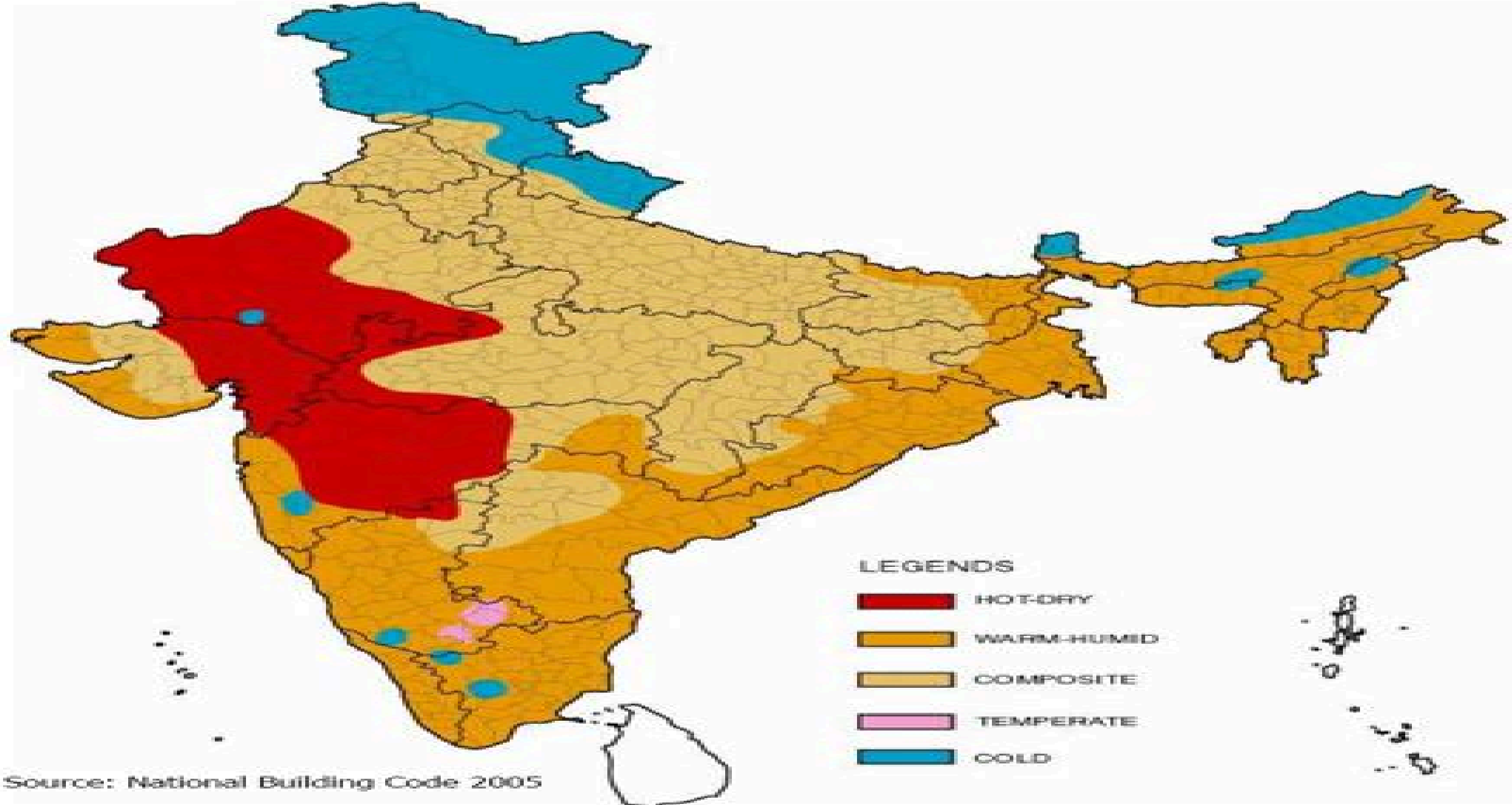


Climate of India

- The climate of India may be broadly described as tropical monsoon type. The term “monsoon” is derived from Arabic word ‘mausim’ which means seasonal reversal in the wind direction


India Climatic Zone Map
Climatic zones (Köppen classification)
Montane
Humid subtropical
Tropical wet and dry
Tropical wet
Semi-arid
Arid
NATIONAL CAPITAL
State Capital
Union Territory Capital
JAMMU & KASHMIR
Srinagar
HIMACHAL PRADESH
Shimla
PUNJAB
Chandigarh
Dehradun
UTTARANCHAL
HARYANA
Delhi
NEW DELHI
RAJASTHAN
Jaipur
UTTAR PRADESH
Lucknow
SIKKIM
Gangtok
ARUNACHAL PRADESH
Itanagar
ASSAM
Guwahati
Shillong
NAGALAND
Kohima
BIHAR
Patna
MEGHALAYA
Imphal
MANIPUR
Gandhinagar
MADHYA PRADESH
JHARKHAND
Aizwal
Agartala
GUJARAT
Bhopal
Ranchi
WEST BENGAL
TRIPURA
MIZORAM
KOLKATA
CHHATTISGARH
Diu
Raipur
ORISSA
Daman & Diu
Silvasaa
Dadra & Nagar Haveli
MAHARASHTRA
Bhubaneshwar
MUMBAI
Hyderabad
ANDHRA PRADESH
Pondicherry (Yanam)
Panaji
GOA
KARNATAKA
Bangalor
CHENNAI
Pondicherry (Mahe)
Pondicherry (Puducherry)
Lakshadweep Islands
Kavaratti
TAMIL NADU
Pondicherry (Karaikal)
KERALA
Andaman & Nicobar Islands
Port Blair
Thiruvananthapuram

Climate Zone Map Of India



Source: National Building Code 2005

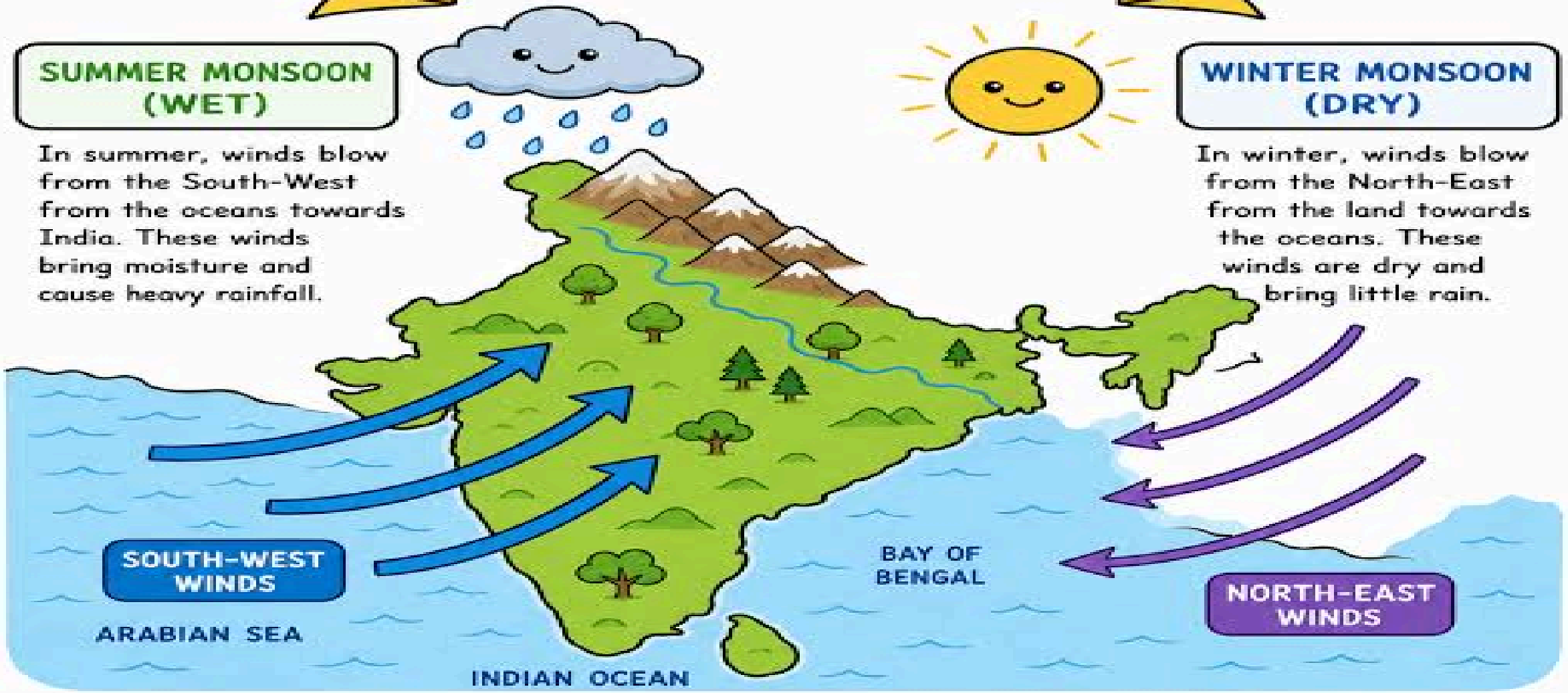
THE MONSOON IN INDIA

SUMMER MONSOON (WET)

In summer, winds blow from the South-West from the oceans towards India. These winds bring moisture and cause heavy rainfall.

WINTER MONSOON (DRY)

In winter, winds blow from the North-East from the land towards the oceans. These winds are dry and bring little rain.



SOUTH-WEST WINDS = WET MONSOON
Bring moisture from the oceans and cause heavy rainfall.

NORTH-EAST WINDS = DRY MONSOON
Blow from the land, are dry and bring little rain.

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- Intense Heating of the Land (May–June):During summer, the Indian subcontinent heats up intensely under the direct sun.
- Land heats up quickly, causing the air above it to warm up, expand, and rise.
- Moisture and Rainfall: It is because these winds travel thousands of kilometers over warm ocean waters, they absorb massive amounts of moisture.

When they hit India's landmass and mountain ranges (like the Western Ghats and Himalayas), the air rises, cools down, forms clouds, and pours down as monsoon rain.

The "Coriolis" Bend: As these ocean winds cross the Equator, the Earth's rotation bends them toward the right (due to the Coriolis force). This curves their direction so that they approach the Indian subcontinent

- This creates a massive low-pressure area over Northern and Central India. Cooler Oceans with High Pressure: At the same time, the surrounding Indian Ocean and Arabian Sea remain much cooler than the land.
- The air above the cooler water is dense and heavy, creating a high-pressure area.
- Nature Balances the Pressure: Air always flows from high pressure to low pressure. Driven by this pressure imbalance, cool, heavy winds start blowing from the Indian Ocean towards the Indian landmass

The Indian Meteorological Department (IMD) designates four official seasons: 1. winter (From December to early April); 2. Summer or Pre-monsoon season (April to July in north-western India); 3. Monsoon or Rainy season (June-September); 4. Post-monsoon season (October-December). But traditionally, Indian note six seasons, each about two month long.

- India’s climate is affected by two seasonal winds — the north-east monsoon and the south-west monsoon. The north-east monsoon commonly known as winter monsoon blows from land to sea whereas south-west monsoon known as summer monsoon blows from sea to land after crossing the Indian Ocean, the Arabian Sea and the Bay of Bengal.

- The south-west monsoon brings most of the rainfall during the year in the country. It is now possible to make forecast about the monsoon rains successfully with developed models.

- 1. Tropical location, the presence of the Himalaya and Indian Ocean defines the special characteristics of Indian climate, but, it's broadly the tropical monsoon type. The region to the South of Tropic of Cancer experiences tropical climate, whereas the region of the North of the Tropic of Cancer experiences warm temperate climate.

- 2. The Northern Himalayan Range separates India from the rest of the India and hence, protects India from the bitterly cold and dry winds of Central Asia during winter. Moreover, it acts as a physical barrier for the rain bearing South-West monsoon.

- 3. India's large size also has significant impact on its climatic pattern, the areas near to the sea experience maritime climate, whereas the areas away from the sea experience continental climate.

- 4. Monsoon is the most dominating factor of Indian climate; it is the seasonal wind circulation, which completely reverses their directions of flow with the change in season.

- 5. The South-West summer monsoon from the Arabian Sea and the Bay of Bengal bring rainfall to the entire country. Besides, the North-East winter monsoon, travelling from land to sea causes rainfall along the Coromandal coast after getting moisture from the Bay of Bengal.

- 6. The Arabian branch of monsoon is more powerful than the Bay of Bengal branch. The onset of monsoon takes place first in Andaman and Nicobar Island and in Kerala coast in the Mainland of India.

- 7. Upper air circulation, the jet stream influences the monsoon as well as the climate in India. The easterly jet stream of the summer season helps with the onset of the South-West monsoon. The South branch of the jet stream in the winter season intensifies high pressure centres over the North-Western India.

- 8. Jet stream theory: It explains that the Westerly jet stream during the winter season bifurcated by the Himalayan ranges and its Southern branch located along the Southern slopes of the Himalayas intensifies the high pressure centres over the North-Western India in winter, but shift to the North with the onset of the monsoon. This periodic movement of the sub-tropical jet stream provides a useful indication of the onset and subsequent withdrawal of the monsoon.

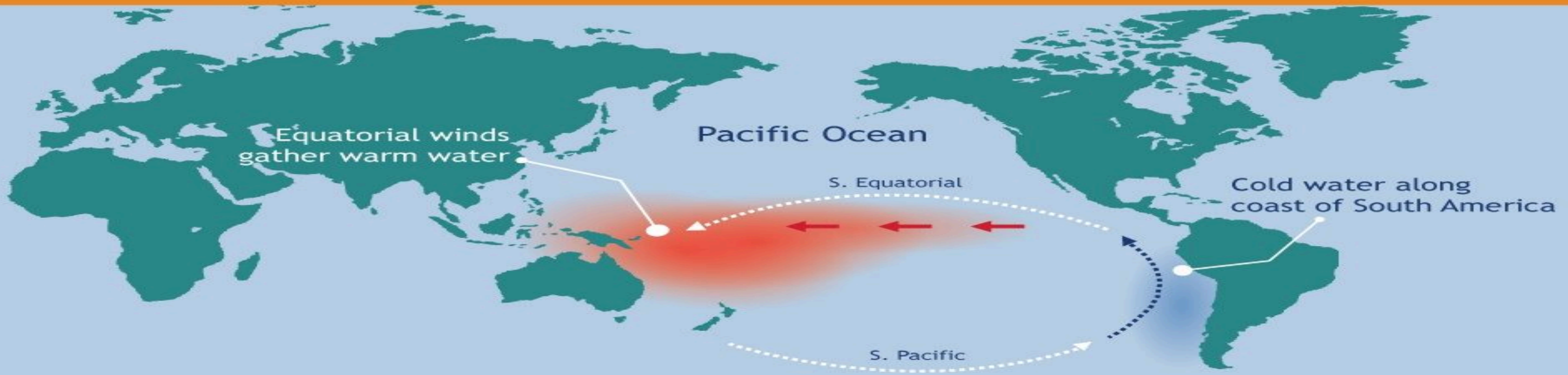
- 9. The tropical cyclones generated in Arabian" Sea and Bay of Bengal during the South-West monsoon and the retreating monsoon seasons, influence the weather conditions of the Peninsular India. The Eastern coast is more prone to tropical cyclone as the number of depressions originated in the Bay of Bengal is more as compared to that of in the Arabian Sea

- In the Western coast, Gujarat and Northern part of the Maharashtra are more affected as compared to the Southern part. But during the period of El-Nino, the Modoki Western coast gets more cyclones than eastern coast in India and during El-Nino period, Eastern coast gets unusually more cyclones.

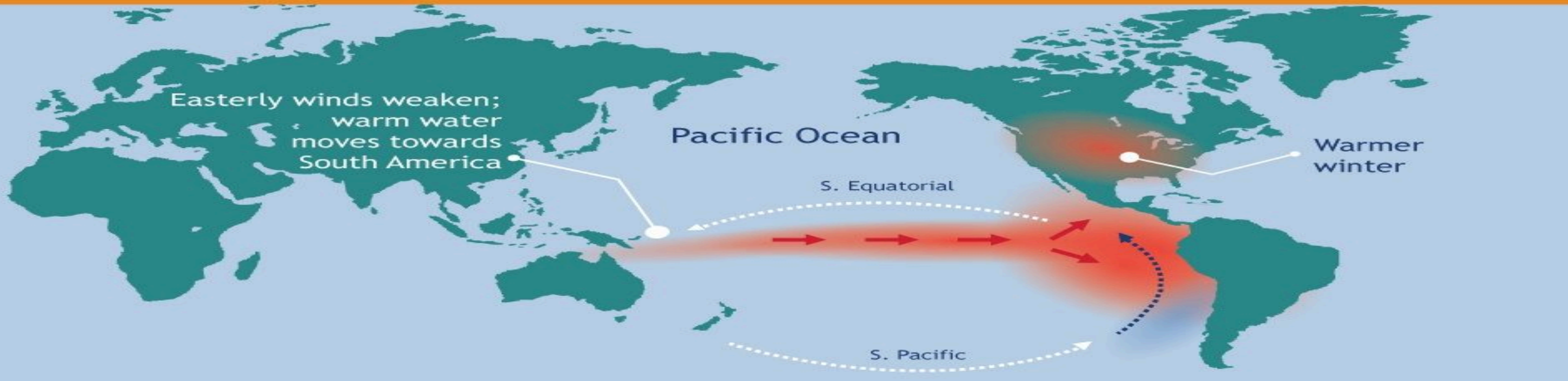
- El Nino is a climate pattern that describes the unusual warming of surface waters in the eastern tropical Pacific Ocean. El Nino is the “warm phase” of a larger phenomenon called the El Nino-Southern Oscillation (ENSO). La Nina, the “cool phase” of ENSO, is a pattern that describes the unusual cooling of the region’s surface waters. El Nino and La Nina are considered the ocean part of ENSO, while the Southern Oscillation is its atmospheric changes.

- El Nino is an occasional narrow warm ocean current appearing along the Peru coast in December replacing the Peru or Humboldt cold ocean current flowing over the region in normal years. The appearance of El-Nino anomaly reverses the thermal condition of the ocean surface (warm condition over the Eastern Pacific, Peru and cold in Western Pacific, Indonesia and also the lower atmospheric wind circulation that is conducive for weak monsoon and characterized with deficient rainfall and drought in India.

NORMAL YEAR



EL NIÑO YEAR



Source: Loomis Sayles depiction based on source data from National Oceanic and Atmospheric Administration (NOAA), National Aeronautics and Space Administration (NASA) and various media reports.

- 10. The Western disturbances influence the winter weather conditions over most of the Northern plains and Western Himalayan regions. La Nina, which is the complete reversal of the El Nino, is the harbinger for the heavy monsoon showers in India.

- 11. El Nino is an occasional narrow warm ocean current appearing along the Peru coast in December replacing the Peru or Humboldt cold ocean current flowing over the region in normal years.

- The appearance of El-Nino anomaly reverses the thermal condition of the ocean surface (warm condition over the Eastern Pacific, Peru and cold in Western Pacific, Indonesia and also the lower atmospheric wind circulation that is conducive for weak monsoon and characterized with deficient rainfall and drought in India.

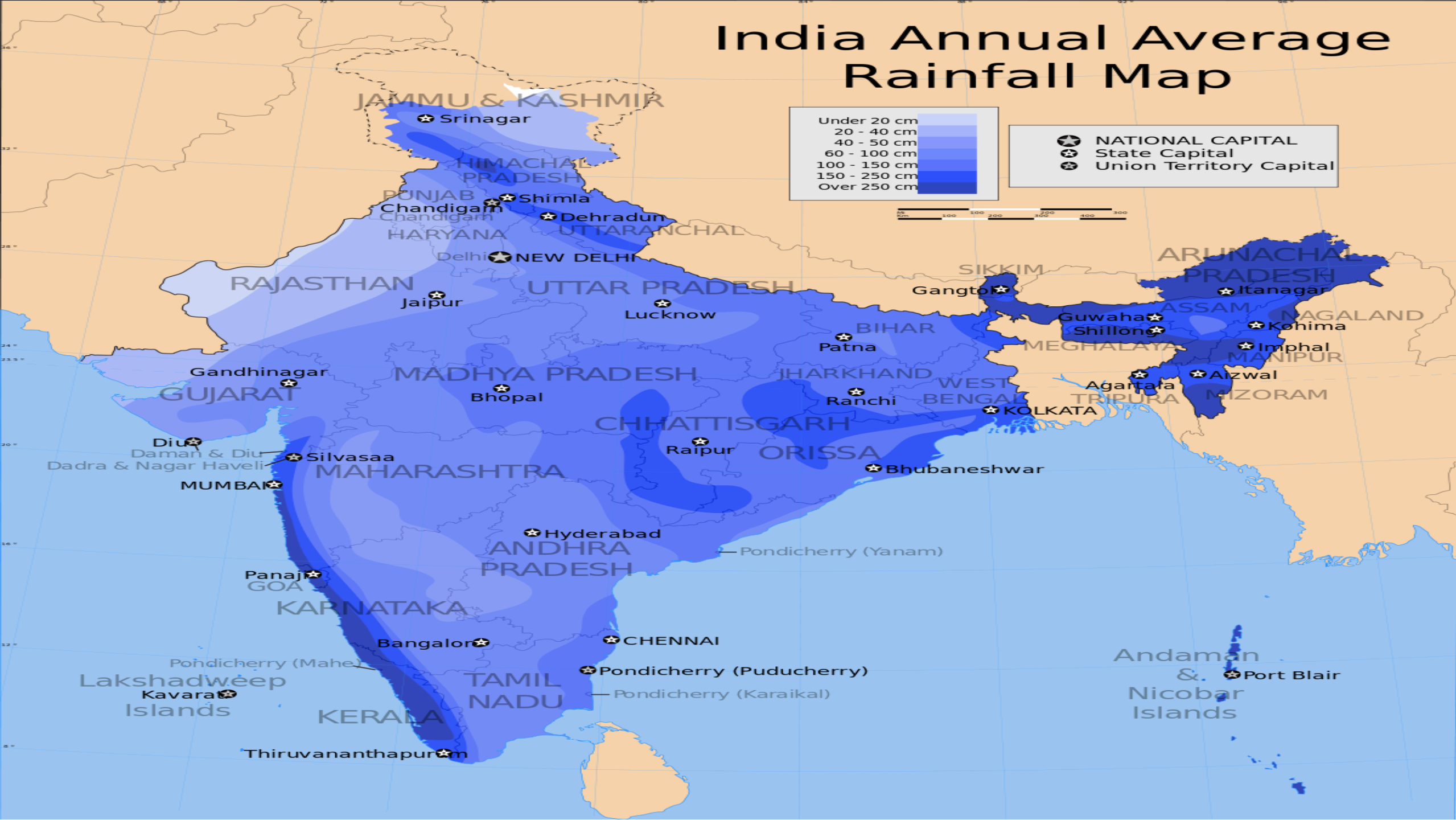
- 12. It has been noticed that whenever the surface pressure is high over the Pacific, the pressure over the Indian Ocean is low and vice-versa; this seesaw pattern of meteorological changes is named 'Southern Oscillation'. There is a close relationship between the appearance of the El-Nino and the negative Southern oscillation and this combined effect is called an ENSO event. According to incidence of rainfall, India is classified into 4 broad categories.

- (I) Areas of very high rainfall: Areas receiving 200 cm and above rainfall includes North-East of India and the Western coast from Thiruvananthapuram to Mumbai.

- (II) Areas of high rainfall: The areas receiving 100-200 cm includes Eastern slopes of Western Ghats, major parts of Northern Plain and East coasts.

- (III) Areas of low rainfall: The areas receiving rainfall between 50-100 cm includes large part of Gujarat, Eastern Rajasthan etc.

- (IV) Areas of very low rainfall: Areas receiving rainfall below 50 cm, includes arid and semi-arid areas of Western Rajasthan, Kutch and most of the region of Ladakh, Jammu and Kashmir.


India Annual Average Rainfall Map
Under 20 cm
20 - 40 cm
40 - 50 cm
60 - 100 cm
100 - 150 cm
150 - 250 cm
Over 250 cm
NATIONAL CAPITAL
State Capital
Union Territory Capital
JAMMU & KASHMIR
Srinagar
HIMACHAL PRADESH
Shimla
PUNJAB
Chandigarh
Dehradun
UTTARANCHAL
HARYANA
Delhi
NEW DELHI
RAJASTHAN
Jaipur
UTTAR PRADESH
Lucknow
SIKKIM
Gangtok
ARUNACHAL PRADESH
Itanagar
ASSAM
Guwahati
Shillong
NAGALAND
Kohima
BIHAR
Patna
MEGHALAYA
Imphal
MANIPUR
Aizwal
MIZORAM
Agartala
TRIPURA
WEST BENGAL
KOLKATA
Gandhinagar
GUJARAT
MADHYA PRADESH
Bhopal
JHARKHAND
Ranchi
CHHATTISGARH
Raipur
ORISSA
Bhubaneshwar
Diu
Daman & Diu
Dadra & Nagar Haveli
Silvasaa
MAHARASHTRA
MUMBAI
Hyderabad
ANDHRA PRADESH
Pondicherry (Yanam)
Panaji
GOA
KARNATAKA
Bangalore
CHENNAI
Pondicherry (Puducherry)
Pondicherry (Mahe)
Pondicherry (Karaikal)
TAMIL NADU
Lakshadweep Islands
Kavaratti
KERALA
Thiruvananthapuram
Andaman & Nicobar Islands
Port Blair

- The Monsoon climate is a characteristics property of south and south-east Asia and has influence over the economic activities of these regions. The Indian agriculture is considered a gamble against monsoon because agricultural activities over almost all parts of the India are very much dependent up on the monsoon rainfall. In fact, monsoon is the axis around which the Indian economy revolves.